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#include <iostream>
#include <cstdlib>
#include <ctime>
using namespace std;

const int max_users = 1000;

int main() {
    string enter_name, name[max_users] = { "Sadeed" };
    int enter_cnic, cnic[max_users] = { 123 };
    int enter_accnum, accnum, account_number[max_users] = { 111 };
    int access, enter_pin, pin[max_users] = { 9999 };
    long int withdraw, deposit, savings[max_users] = { 1200 },
defaults[max_users] = { 500 }, current[max_users] = { 1500 };
    int option;
    int userindex = -1;

    start:
        cout<<"1 Account Detail \n2 Open an Account \n3 Close an Account
\n4 Show Balance \n5 Deposit Amount \n6 Withdraw Amount \n7 Exit \nSelect
Option: ";
        cin>>option;
        cout<<endl;

        switch(option)
        {
            case 1:
                start1:
                    userindex = -1;
                    cout<<"Enter your CNIC Number: ";
                    cin>>enter_cnic;

                    for( int i=0; i < max_users; i++ )
                    {
                        if ( enter_cnic == cnic[i] )
                        {
                            userindex = i;
                            break;
                        }
                    }

                    if ( userindex != -1 )
                    {
                        cout<<"Account found!"<<endl<<"Details:"<<endl;
                        cout<<"Account Owner: "<<name[userindex]<<endl;

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        cout<<"Account Number:
"<<account_number[userindex]<<endl;
        cout<<"Account Pin: "<<pin[userindex]<<endl<<endl;
        goto start;
    }
    else
    {
        cout<<"No Record Found!"<<endl<<endl;
        nav1:
            cout<<"1 Back \n2 Main Menu \n3 Exit \nSelect
Option: ";

            cin>>option;
            switch(option)
            {
                case 1:
                    cout<<endl;
                    goto start1;
                    break;
                case 2:
                    cout<<endl;
                    goto start;
                    break;
                case 3:
                    cout<<"Exiting System!";
                    return 0;
                    break;
                default:
                    cout<<"Invalid Selection!"<<endl;;
                    goto nav1;
                    break;
            }
        }
        break;

    case 2:
        start2:
            userindex = -1;
            nam:
                getline(cin, enter_name);
            if (enter_name.empty())
            {
                cout<<"Enter your name: ";
                goto nam;
            }
            cout<<"Enter your CNIC number (3 digits): ";
            cin>>enter_cnic;

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for( int i=0; i < max_users; i++ )
{
    if ( enter_cnic == cnic[i] )
    {
        userindex = i;
        break;
    }
}
if (userindex != -1)
{
    cout<<"An account is already registered to this
CNIC!";

    nav2:
    cout<<"\n\n1 Back \n2 Main Menu \n3 Exit
\nSelect Option: ";

    cin>>option;
    switch(option)
    {
        case 1:
            cout<<endl;
            goto start2;
            break;
        case 2:
            cout<<endl;
            goto start;
            break;
        case 3:
            cout<<"Exiting System!";
            return 0;
            break;
        default:
            cout<<"Invalid Selection!"<<endl;;
            goto nav2;
            break;
    }

}
else
{
    for( int i=0; i < max_users; i++ )
    {
        if( cnic[i] == 0 )
        {

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        name[i] = enter_name;
        cnic[i] = enter_cnic;

        again:
            srand(time(0));
            accnum = rand() % 1000 + 100;
            for( int j=0; j < max_users; j++ )
            {
                if( accnum ==
account_number[j] )

                    goto again;
            }

            account_number[i] = accnum;

        create:
            cout<<"Create a 4 digit pin: ";
            cin>>enter_pin;
            if (enter_pin > 999 && enter_pin < 10000
)

            {
                cout<<"Pin created!";
                pin[i] = enter_pin;
                cout<<endl<<endl;
                goto start;
            }
            else
            {
                cout<<"Please enter a 4 digit
pin!"<<endl<<endl;

                goto create;
            }
            break;
        }
    }
    break;

case 3:
    start3:
        userindex = -1;
        cout<<"Enter Your Account Number: ";
        cin>>enter_accnum;

        for( int i=0; i < max_users; i++ )
        {

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        if( enter_accnum == account_number[i] )
        {
            userindex = i;
            break;
        }
    }

    if( userindex != -1 )
    {
        char sure;
        confirm:
            cout<<"Are you sure you want to delete your
account?(y/n): ";

            cin>>sure;
            if( sure == 'y' || sure == 'Y' )
            {
                name[userindex] = "";
                cnic[userindex] = 0;
                account_number[userindex] = 0;
                pin[userindex] = 0;
                cout<<"Account Deleted!"<<endl<<endl;
                goto start;
            }
            else if( sure == 'n' || sure == 'N' )
            {
                cout<<"Account Not Deleted!"<<endl<<endl;
                goto start;
            }
            else
            {
                cout<<"Invalid Option!"<<endl;
                goto confirm;
            }
        }
    }
    else
    {
        cout<<"No Account Found!";

        nav3:
            cout<<"\n\n1 Back \n2 Main Menu \n3 Exit
\nSelect Option: ";

            cin>>option;

            switch(option)
            {

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        case 1:
            cout<<endl;
            goto start3;
            break;
        case 2:
            cout<<endl;
            goto start;
            break;
        case 3:
            cout<<"Exiting System!";
            return 0;
            break;
        default:
            cout<<"Invalid Selection!"<<endl;;
            goto nav3;
            break;
    }

}

break;

case 4:
    start4:
        userindex = -1;
        cout<<"Enter Account Number: ";
        cin>>enter_accnum;

        for( int i=0; i < max_users; i++ )
        {
            if ( enter_accnum == account_number[i] )
            {
                userindex = i;
                break;
            }
        }
        if ( userindex != -1 )
        {
            retry1:
                cout<<"Enter Account Pin: ";
                cin>>enter_pin;

                if( enter_pin == pin[userindex] )
                {
                    sub_start4:
                        cout<<"1 Current \n2 Default \n3 Savnigs
\nSelect Account: ";

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        cin>>access;

switch(access)
{
    case 1:
        cout<<"Current Account Balance:
"<<current[userindex];

        break;
    case 2:
        cout<<"Default Account Balance:
"<<defaults[userindex];

        break;
    case 3:
        cout<<"Savings Account Balance:
"<<savings[userindex];

        break;
    default:
        cout<<"Invalid Selection!";
}

nav4_1:
    cout<<"\n\n1 Back \n2 Main Menu \n3 Exit
\nSelect Option: ";

    cin>>option;

switch(option)
{
    case 1:
        cout<<endl;
        goto sub_start4;
        break;
    case 2:
        cout<<endl;
        goto start;
        break;
    case 3:
        cout<<"Exiting System!";
        return 0;
        break;
    default:
        cout<<"Invalid Selection!"<<endl;;
        goto nav4_1;
        break;
}
}
else

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        {
            cout<<"Invalid Pin!"<<endl<<endl;
            goto retry1;
        }
    }
    else
    {
        cout<<"No Account Found!"<<endl<<endl;

        nav4_2:
            cout<<"1 Back \n2 Main Menu \n3 Exit

\nSelect Option: ";

            cin>>option;

            switch(option)
            {
                case 1:
                    cout<<endl;
                    goto start4;
                    break;
                case 2:
                    cout<<endl;
                    goto start;
                    break;
                case 3:
                    cout<<"Exiting System!";
                    return 0;
                    break;
                default:
                    cout<<"Invalid Selection!"<<endl;;
                    goto nav4_2;
                    break;
            }
        }
        break;

    case 5:
        start5:
            userindex = -1;
            cout<<"Enter Account Number: ";
            cin>>enter_accnum;
            for( int i=0; i < max_users; i++ )
            {
                if ( enter_accnum == account_number[i] )
                {
                    userindex = i;

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        break;
    }
}
if ( userindex != -1 )
{
    retry2:
        cout<<"Enter Account Pin: ";
        cin>>enter_pin;

        if( enter_pin == pin[userindex] )
        {
            sub_start5:
                cout<<"1 Current \n2 Default \n3 Savnigs
\nSelect Account In Which You Want To Deposit Amount: ";
                cin>>access;
                cout<<"Enter Amount That You Want To
Deposit: ";

                cin>>deposit;

                switch(access)
                {
                    case 1:
                        current[userindex] += deposit;
                        cout<<"Current Account Balance:
"<<current[userindex];

                        break;
                    case 2:
                        defaults[userindex] += deposit;
                        cout<<"Default Account Balance:
"<<defaults[userindex];

                        break;
                    case 3:
                        savings[userindex] += deposit;
                        cout<<"Savings Account Balance:
"<<savings[userindex];

                        break;
                    default:
                        cout<<"Invalid Selection!";
                }

            nav5_1:
                cout<<"\n\n1 Back \n2 Main Menu \n3 Exit
\nSelect Option: ";

                cin>>option;

                switch(option)

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        {
            case 1:
                cout<<endl;
                goto sub_start5;
                break;
            case 2:
                cout<<endl;
                goto start;
                break;
            case 3:
                cout<<"Exiting System!";
                return 0;
                break;
            default:
                cout<<"Invalid Selection!"<<endl;;
                goto nav5_1;
                break;
        }
    }
    else
    {
        cout<<"Invalid Pin!"<<endl<<endl;
        goto retry2;
    }
}
else
{
    cout<<"No Account Found!"<<endl<<endl;
    nav5_2:
        cout<<"1 Back \n2 Main Menu \n3 Exit
\nSelect Option: ";

        cin>>option;

        switch(option)
        {
            case 1:
                cout<<endl;
                goto start5;
                break;
            case 2:
                cout<<endl;
                goto start;
                break;
            case 3:
                cout<<"Exiting System!";
                return 0;
        }
    }
}

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                break;
            default:
                cout<<"Invalid Selection!"<<endl;;
                goto nav5_2;
                break;
        }
    }
    break;

case 6:
    start6:
    userindex = -1;
    cout<<"Enter Account Number: ";
    cin>>enter_accnum;
    for( int i=0; i < max_users; i++ )
    {
        if ( enter_accnum == account_number[i] )
        {
            userindex = i;
            break;
        }
    }
    if ( userindex != -1 )
    {
        retry3:
        cout<<"Enter Account Pin: ";
        cin>>enter_pin;

        if( enter_pin == pin[userindex] )
        {
            sub_start6:
            cout<<"1 Current \n2 Default \n3 Savnigs
\nSelect Account From Which You Want To Withdraw Amount: ";
            cin>>access;
            cout<<"Enter Amount That You Want To
Withdraw: ";

            cin>>withdraw;

            switch(access)
            {
                case 1:
                    if( withdraw > current[userindex] )
                    {
                        cout<<"Insufficient
Balance!";
                    }

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        else
        {
            current[userindex] -=
                withdraw;
            cout<<"Current Account
            Balance: "<<current[userindex];
        }
        break;
    case 2:
        if( withdraw > defaults[userindex]
        )
        {
            cout<<"Insufficient
            Balance!";
        }
        else
        {
            defaults[userindex] -=
                withdraw;
            cout<<"Default Account
            Balance: "<<defaults[userindex];
        }
        break;
    case 3:
        if( withdraw > savings[userindex] )
        {
            cout<<"Insufficient
            Balance!";
        }
        else
        {
            savings[userindex] -=
                withdraw;
            cout<<"Savings Account
            Balance: "<<savings[userindex];
        }
        break;
    default:
        cout<<"Invalid Selection!";
    }

    nav6_1:
        cout<<"\n\n1 Back \n2 Main Menu \n3 Exit
        \nSelect Option: ";
        cin>>option;

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        switch(option)
        {
            case 1:
                cout<<endl;
                goto sub_start6;
                break;
            case 2:
                cout<<endl;
                goto start;
                break;
            case 3:
                cout<<"Exiting System!";
                return 0;
                break;
            default:
                cout<<"Invalid Selection!"<<endl;;
                goto nav6_1;
                break;
        }
    }
    else
    {
        cout<<"Invalid Pin!"<<endl<<endl;
        goto retry3;
    }
}
else
{
    cout<<"No Account Found!"<<endl<<endl;
    nav6_2:
        cout<<"1 Back \n2 Main Menu \n3 Exit
\nSelect Option: ";

        cin>>option;

        switch(option)
        {
            case 1:
                cout<<endl;
                goto start6;
                break;
            case 2:
                cout<<endl;
                goto start;
                break;
            case 3:
                cout<<"Exiting System!";

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        return 0;
        break;
    default:
        cout<<"Invalid Selection!"<<endl;;
        goto nav6_2;
        break;
    }
}
break;

case 7:
    cout<<"Exiting System!";
    break;

default:
    cout<<"Invalid Selection!"<<endl<<endl;
    goto start;
}

return 0;
}

```